# HRISHIN DEBNATH

Student Researcher | Multilingual AI and Language Models

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## PROFILE

Higher-secondary science student in Dhaka. I am building a Bangla version of a medical exam question set to test whether language models answer worse in Bangla than in English. I am also the sole author of two papers on Dhaka's air quality, both under review. I am looking for a professor to guide my next study on language disparity in LLMs and on image-generation models.

**Research interests:** language disparity in LLMs (Bangla and other low-resource languages against English), LLM evaluation, machine translation quality, image-generation models

**Next study:** go deeper into language disparity in LLMs, building on the translated MedQA set, and extend it to image-generation models: does a model produce worse images from a Bangla prompt than from the same prompt in English?

## RESEARCH

**Catching Translation Errors in Bangla Medical Exam Questions (working title)** **Jul 2026 – Present**

*Researcher, mentored study at CreateBud Inc. In preparation*

- Building a Bangla version of the 12,000-question MedQA exam set; in a 500-question pilot of 5 open models, the revised prompt cut flagged translations for the chosen model from 399 to 100.
- Tested the error checks with planted mistakes: they caught 50 of 50 swapped answer options and 36 of 50 tenfold number changes.

**Next-Day PM2.5 Forecasting in Dhaka, Bangladesh, from a Single Reference-Grade Monitor: A Provenance-Audited Gradient-Boosting Approach Using ERA5 Reanalysis and Open Satellite Fire Data** **Sep 2026**

*Sole author. Submitted to Air Quality, Atmosphere & Health (under review)* | **Code**

- Beat the persistence baseline by 5.22 µg/m³ RMSE (30.80 vs 36.02, $R^2$ 0.795) on a 12-month holdout with XGBoost, using 248,953 hourly rows merged from OpenAQ, ERA5, NASA POWER and NASA FIRMS.
- Checked the result with walk-forward testing and an independent LightGBM model tuned with Optuna.

**Does Dhaka's Air Get Cleaner When the City Empties? A Nine-Year Natural Experiment Using Eid Holidays and Reference-Grade PM2.5 Data** **Sep 2026**

*Sole author. Columbia Junior Science Journal (under review)* | **Code and data**

- Found PM2.5 falls 36% on average when Dhaka empties for Eid (95% CI 22% to 47%), across 16 holidays and 75,325 hourly U.S. Embassy readings, using an event study and 2,000 placebo runs.

## RESEARCH EXPERIENCE

**CreateBud Inc.** **Remote, part-time**

*Research Intern* *Jul 2026 – Present*

- Running the Bangla medical translation study (see Research) with a CreateBud mentor.
- Tested CreateBud's pre-release co-working platform and reported 6+ bugs in its project, team and task features.

## TECHNICAL SKILLS

**Programming:** Python (pandas, NumPy, SciPy, scikit-learn, statsmodels), R, SQL, DAX; working knowledge of JavaScript and TypeScript

**ML & statistics:** XGBoost, LightGBM, Random Forest, time-series forecasting, walk-forward validation, Optuna, hypothesis testing, event studies

**Data & tools:** PostgreSQL, MySQL, MongoDB, Power BI, Tableau, Excel, Power Query, Power Automate, Git, Jupyter

## SELECTED PROJECTS

**Desk Watch: on-device study-focus app (open source, beta)** | **GitHub** **Electron, TypeScript, MediaPipe**

- Windows app that uses the webcam, fully on the device, to ring an alarm when the user picks up a phone, leaves the desk, looks away or falls asleep; also has a site and app blocker, a planner, a Pomodoro focus mode and a Bangla interface.
- Released as open source (MIT) in October 2026, with a Windows installer and 267 unit tests. Co-written with Claude Code: I wrote about 60% of the code, set the requirements, made the design decisions, recorded the training clips and tested each stage.

**NASA Space Apps Challenge 2026: Baseline (Team Obsidian Prime, in progress)** | **GitHub** **Python, JavaScript**

- Technical lead on a prototype that compares an astronaut's daily health readings with their own normal range and cites the NASA study behind every flag.

## OTHER EXPERIENCE

**VibeWeave (early-stage startup)** **Remote, part-time**
*Market Analyst* *Jun 2026 – Present*

- Built market studies of India's textile industry for five regions (fabric prices, raw-material commodities, listed companies) and a verified database of 26 clothing manufacturers.

**Elevvo Pathways** | **Work on GitHub** **Remote, part-time**
*Data Analyst Intern* *Aug 2025 – Sep 2025*

- Segmented 1,998 retail customers with RFM analysis in Python; 306 "Champions" generated 23.81% of revenue.

**CORViS (student-led non-profit)** | **corvisbd.org** **Dhaka**
*Deputy Director, Research & Development* *Dec 2024 – Present*

- Organised 5+ seminars in the Speaks Academia series by inviting professors and industry professionals, including a panel on careers and AI in economics.

## EDUCATION

**St. Joseph Higher Secondary College, Dhaka** **2026 – 2028 (expected)**
*Higher Secondary Certificate (HSC), Science, English version*

**Cosmo School, Dhaka** **2026**
*Secondary School Certificate (SSC), Science, English version. GPA 5.00 out of 5.00 (1,160 of 1,250 marks)*

**WorldQuant University** | **Credential** **Dec 2025 – Jun 2026**
*Applied Data Science Lab*

- Completed eight end-to-end projects, each passed at 90% or higher: regression, classification, clustering, time-series and GARCH forecasting, A/B testing, and serving models through an API.

## CERTIFICATIONS

**Certificates: Google Data Analytics** (Oct 2024), **Google Business Intelligence** (Nov 2024), **freeCodeCamp Data Analysis with Python** (Jan 2024), **freeCodeCamp Relational Database** (Jul 2024), **freeCodeCamp JavaScript Algorithms and Data Structures** (Sep 2024), **HackerRank SQL (Advanced)** (May 2024), **Cisco Data Analytics Essentials** (Apr 2024), **LinkedIn Advanced Power BI: DAX** (Sep 2024)

**Job simulations (Forage): Quantium Data Analytics** (Aug 2025), **Tata Data Visualisation** (Jul 2024), **Accenture Data Analytics and Visualization** (Jun 2024)